# ISA 401: Business Intelligence & Data Visualization

## 15: Fundamentals of Data Visualization

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Automated Scheduler for Office Hours

Fall 2026

# Before We Start

**1 Join today's Zoom**

Use the Zoom link in the **Week 0 Module** on Canvas. **Mute** your mic and **turn off** your video.
There is no R today. Keep a piece of paper and a pen ready: one activity asks you to sketch a chart.

**2 Today's plan**

**35 min.** Graphical excellence and graphical integrity.
**10 min.** Data-ink and chartjunk.
**18 min.** Data types and visual encoding.
**7 min.** Color.
**Monday:** we turn these principles into ggplot2 code.

# Where We Are

**ISA 401**

↓

| Phase I | Phase II (we are here) | Phase III |
|---|---|---|
| **Data Acquisition & Transformation** | **Data Visualization & Communication** | **Exploratory Data Mining & Project Synthesis** |
| Aug 24 to Oct 12 (ends with Exam 01) | Oct 14 to Nov 16 (ends with Exam 02) | Nov 18 to Dec 2 |

**Phase II starts today.** We collected and cleaned data in Phase I. Now we decide how to show it.

Phase names and dates from the course schedule on Canvas.

# Learning Objectives

By the end of today, you should be able to:

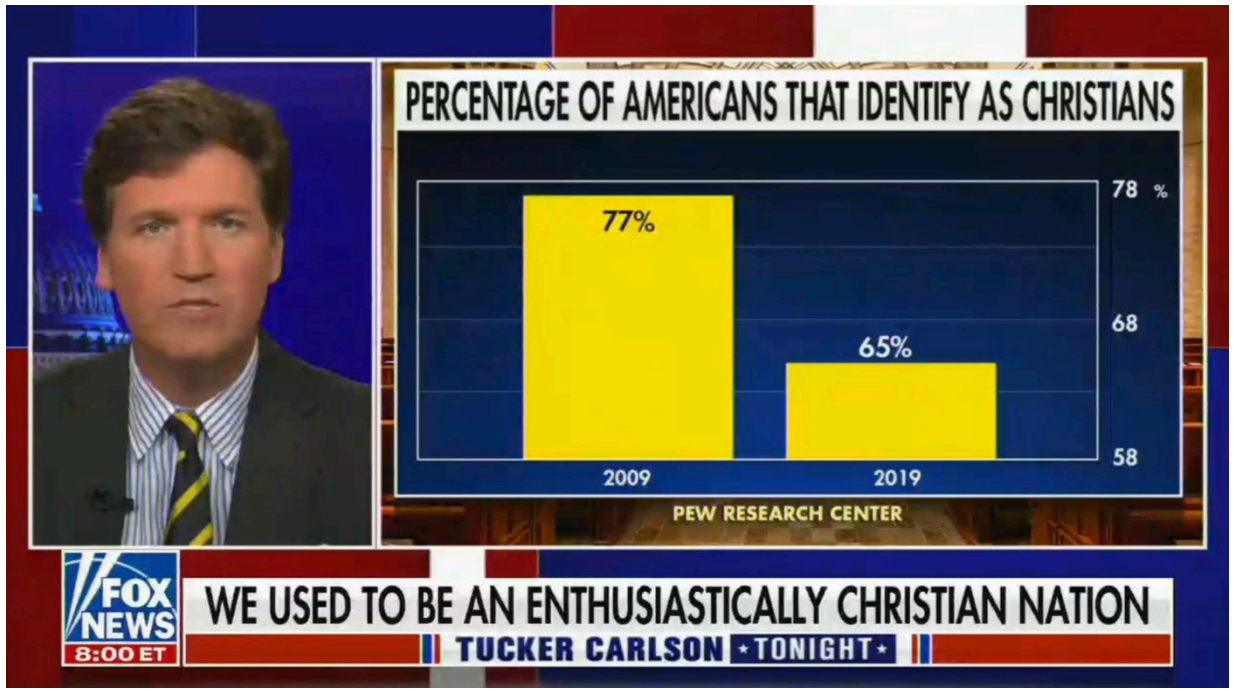
- **Explain** Tufte's principles of graphical excellence.
- **Select** appropriate visual encodings based on data types.
- **Apply** color theory and accessibility guidelines.
- **Identify** and critique misleading visualizations.

# Part 1: Graphical Excellence

# Terrible Charts

05:00

Task | Russia's Defense Budget | White House Growth | Fox News | Discussion




- 77% bar and 65% bar isn't that much of a drastic difference, the gap between the 2 bars seems larger than it is. just 12% difference.
- since th ey asix didn't start at 0, the 77% appears ~ 2-2.5X larger than the 65%(obvisouly mathmatically wrong)

Source: shown in this Fox News clip.

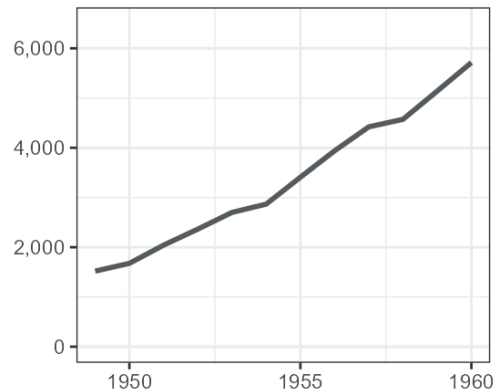
# What Should Graphs Do?

1. **Show the data.**
2. Lead the viewer to think about the **substance** rather than something else.
3. Avoid **distorting** what the data have to say.
4. Present **many numbers in a small space**.
5. Make **large datasets coherent**.
6. Encourage the eye to **compare** different pieces of data.
7. Reveal the data at **several levels of detail**, from a broad overview to the fine structure.
8. Serve a **clear purpose**: description, exploration, tabulation, or decoration.
9. Be closely **integrated with the statistical and verbal descriptions** of the data.

**9. Integrate with the statistics and the text.** The title and labels say what the graph shows.

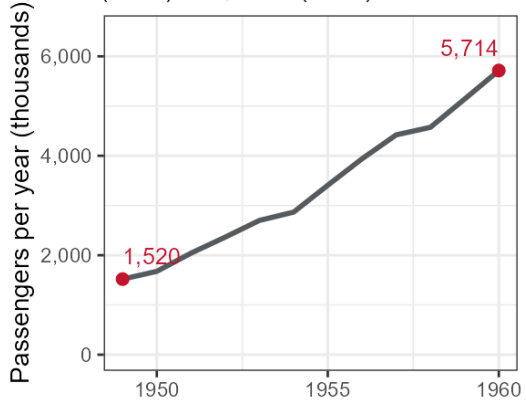
**Less effective: no message**

Figure 1




**Better: title states the finding**

Airline passengers grew from 1,520K (1949) to 5,714K (1960)




Back | Step | Play | Reset | 0.5x | 1x | 2x

9 / 9

Source: Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). Cheshire, CT: Graphics Press, p. 13.

# Show the Data: Anscombe's Quartet

**In a seminal paper, Anscombe wrote:**

> **Few of us escape being indoctrinated with these notions:**
>
> - numerical **calculations are exact, but graphs are rough**;
> - for any particular kind of **statistical data there is just one set of calculations constituting a correct statistical analysis**;
> - performing **intricate calculations is virtuous**, whereas **actually looking at the data is cheating**.

He then argued that

> a computer should **make both calculations and graphs**. Both sorts of output should be studied; each will contribute to understanding.

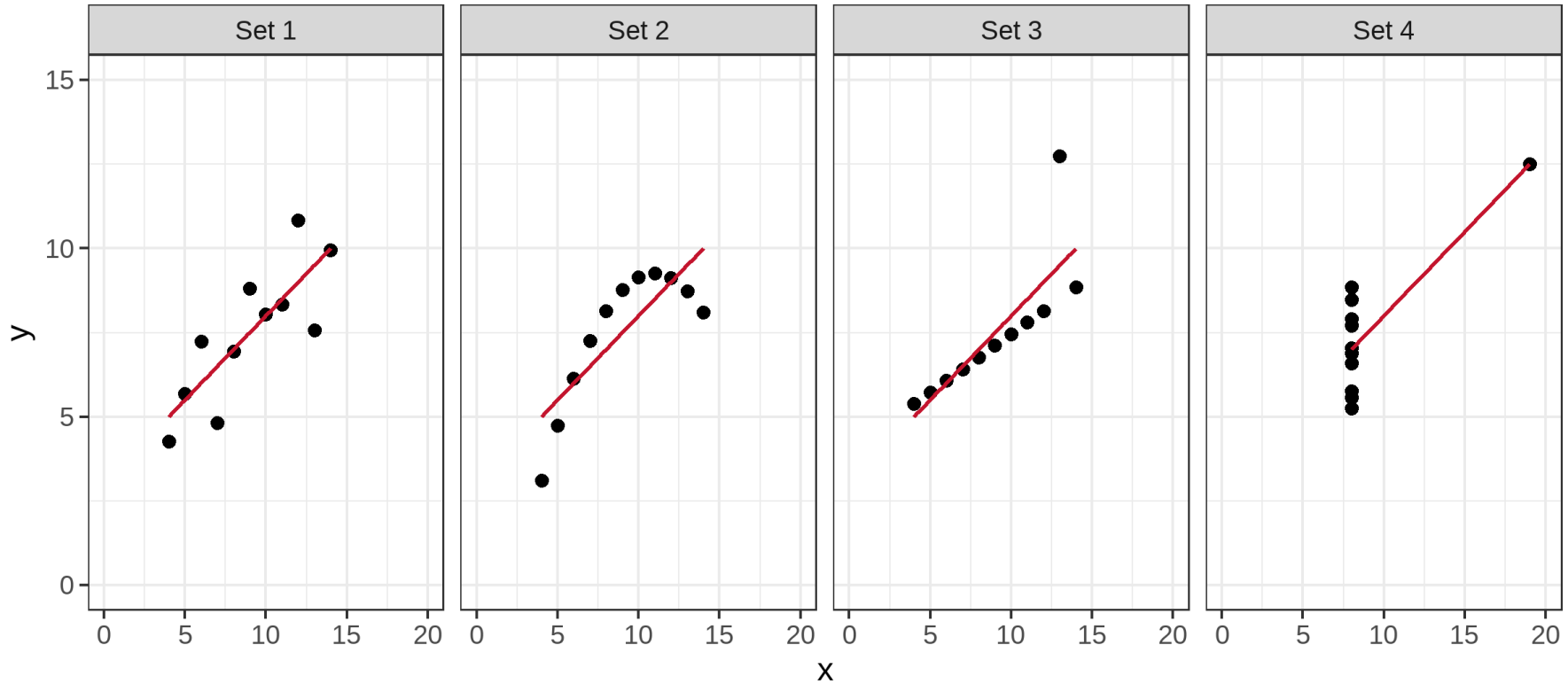
Source: Anscombe, F. J. (1973). Graphs in Statistical Analysis. *The American Statistician*, 27(1), 17-21 (PDF).

# Show the Data: Anscombe's Statistics

Summary statistics | The 44 points

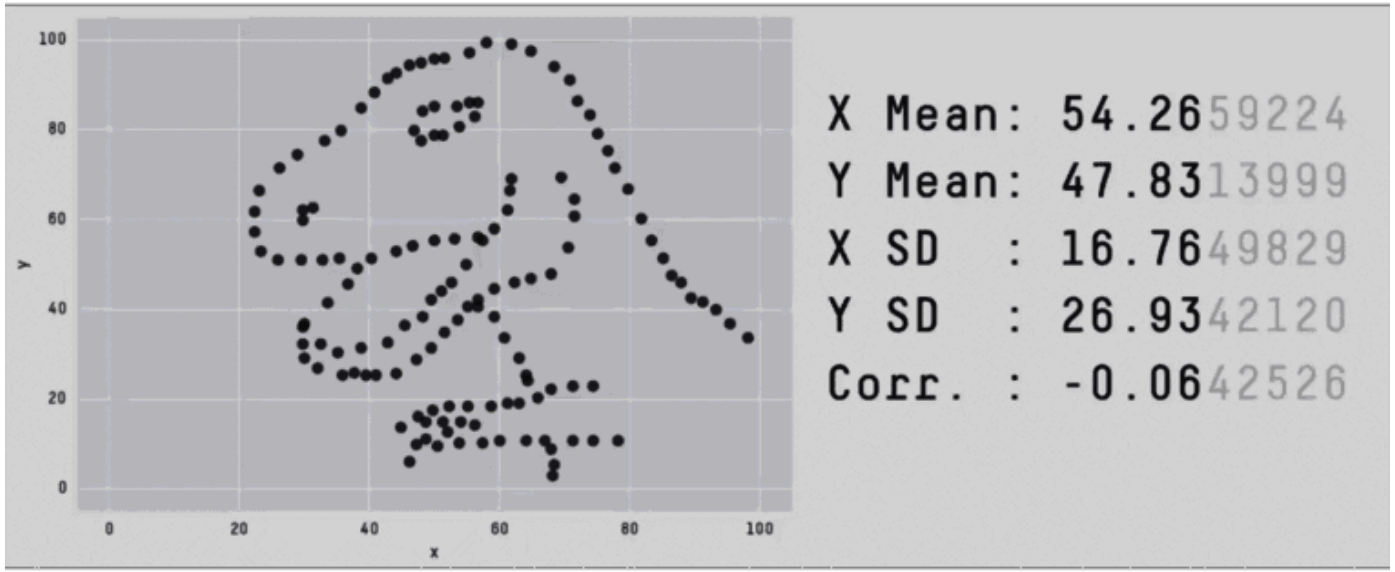
| x1 | y1 | x2 | y2 | x3 | y3 | x4 | y4 |
|---|---|---|---|---|---|---|---|
| 10 | 8.04 | 10 | 9.14 | 10 | 7.46 | 8 | 6.58 |
| 8 | 6.95 | 8 | 8.14 | 8 | 6.77 | 8 | 5.76 |
| 13 | 7.58 | 13 | 8.74 | 13 | 12.74 | 8 | 7.71 |
| 9 | 8.81 | 9 | 8.77 | 9 | 7.11 | 8 | 8.84 |
| 11 | 8.33 | 11 | 9.26 | 11 | 7.81 | 8 | 8.47 |
| 14 | 9.96 | 14 | 8.10 | 14 | 8.84 | 8 | 7.04 |
| 6 | 7.24 | 6 | 6.13 | 6 | 6.08 | 8 | 5.25 |
| 4 | 4.26 | 4 | 3.10 | 4 | 5.39 | 19 | 12.50 |
| 12 | 10.84 | 12 | 9.13 | 12 | 8.15 | 8 | 5.56 |
| 7 | 4.82 | 7 | 7.26 | 7 | 6.42 | 8 | 7.91 |
| 5 | 5.68 | 5 | 4.74 | 5 | 5.73 | 8 | 6.89 |

# Show the Data: Anscombe's Scatterplots




**The four datasets have the same summary statistics but different patterns.** Set 1 is a noisy line, Set 2 is a curve, Set 3 is a line with one outlier, and in Set 4 one point sets the slope. The fitted red line is the same in all four.

# Show the Data: The Datasaurus




Source: Matejka, J. and Fitzmaurice, G. (2017). Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing. *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (project page).

# Substance: Playfair's Cities

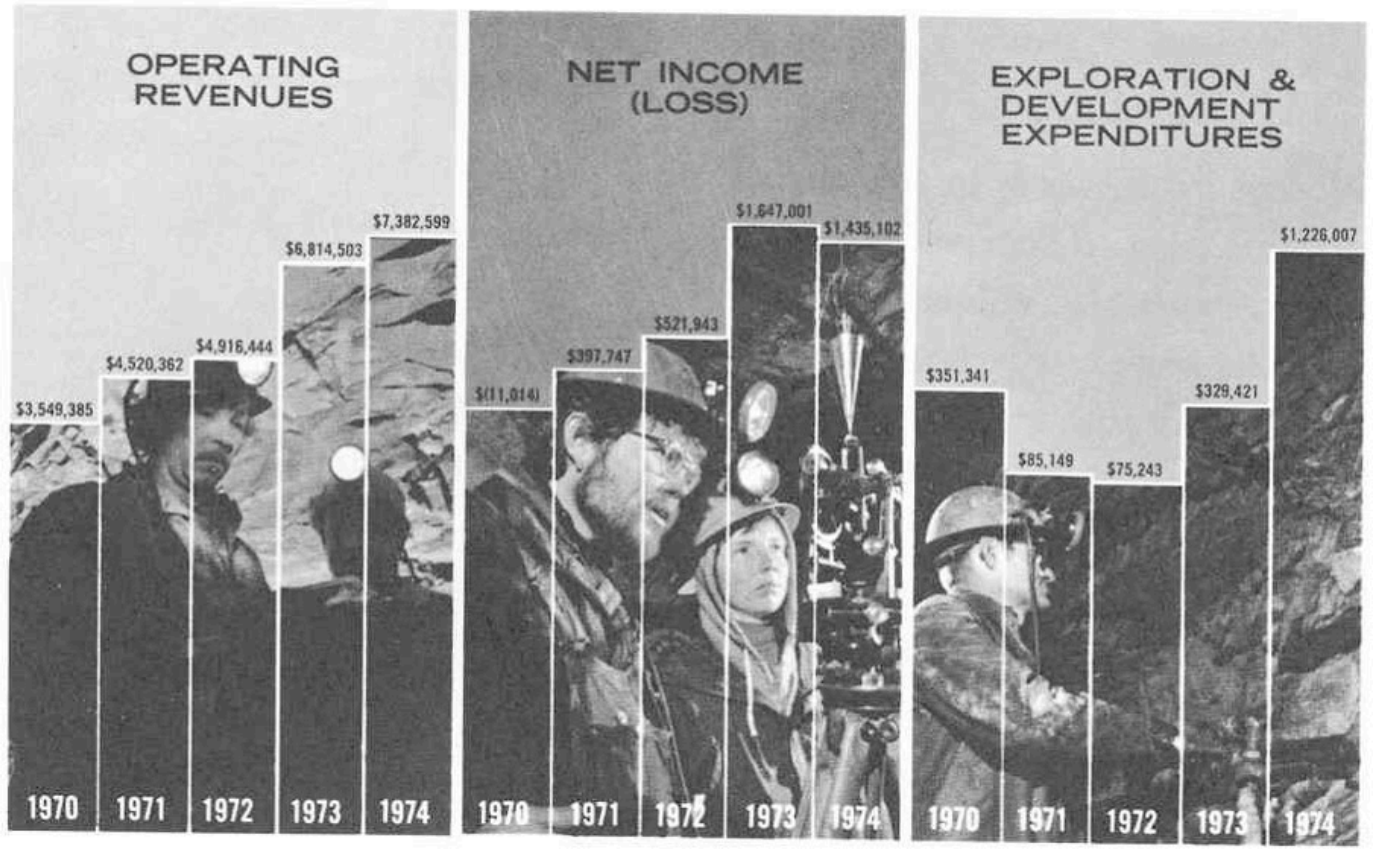
05:00

Task | Your Solution | My Solution | Discussion

▶ Show my solution

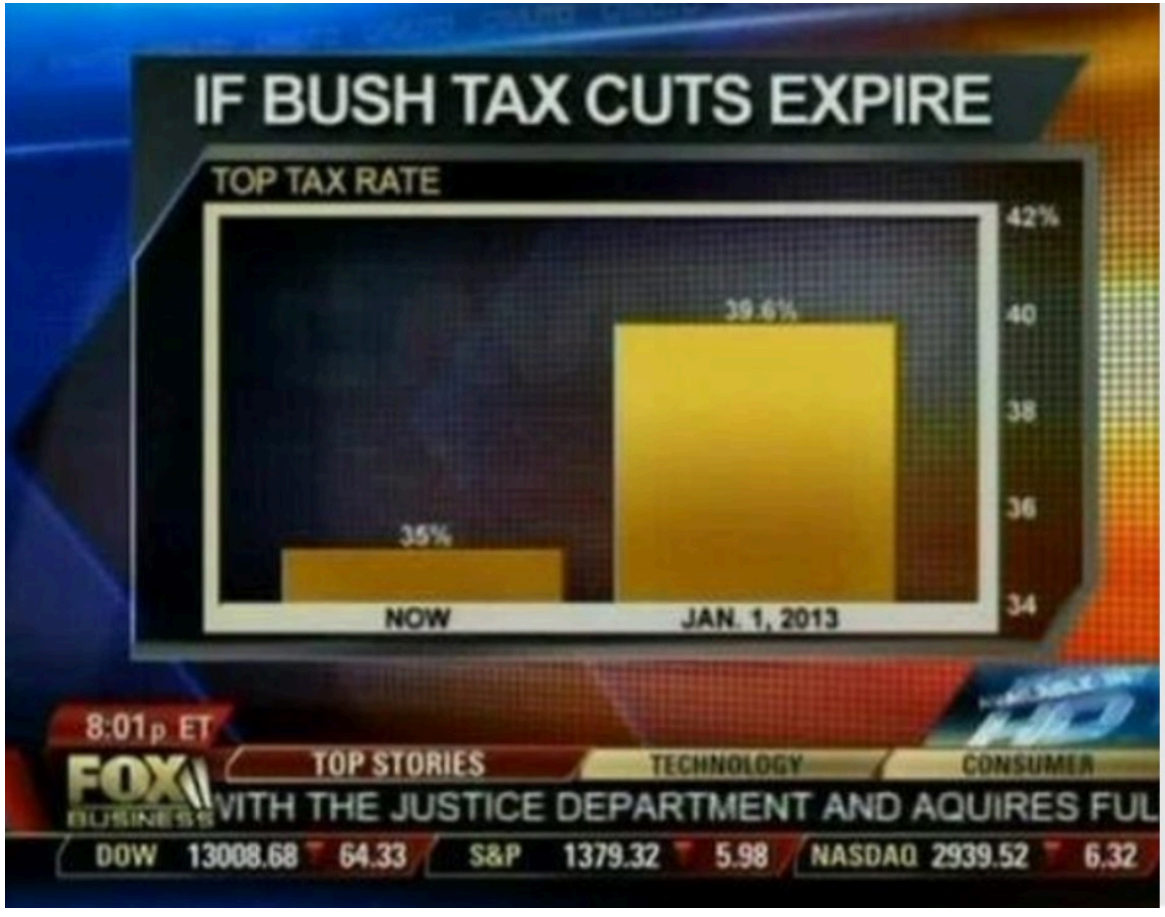
Password [ ] Unlock

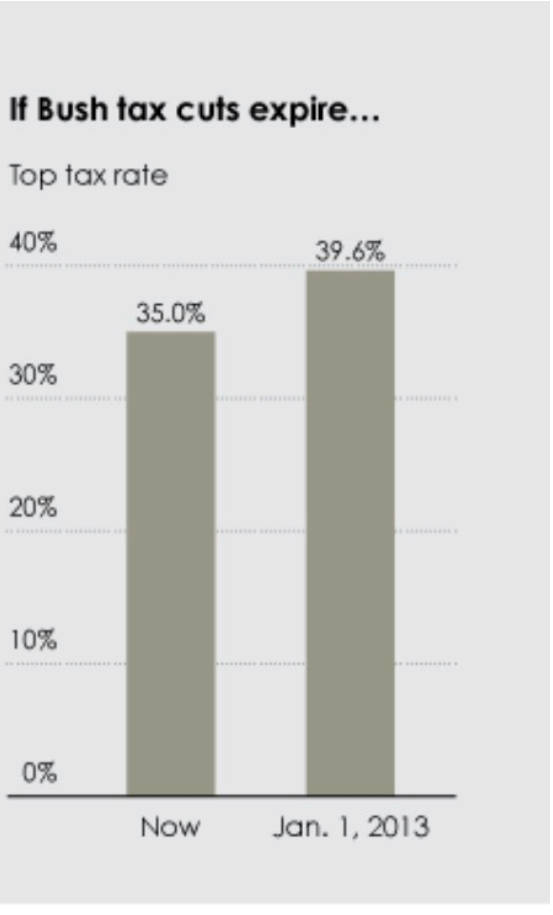
# Avoid Distortion: A Chart from Tufte's Book




Source: Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). Graphics Press, p. 54.

# Avoid Distortion: A Fox Business Chart

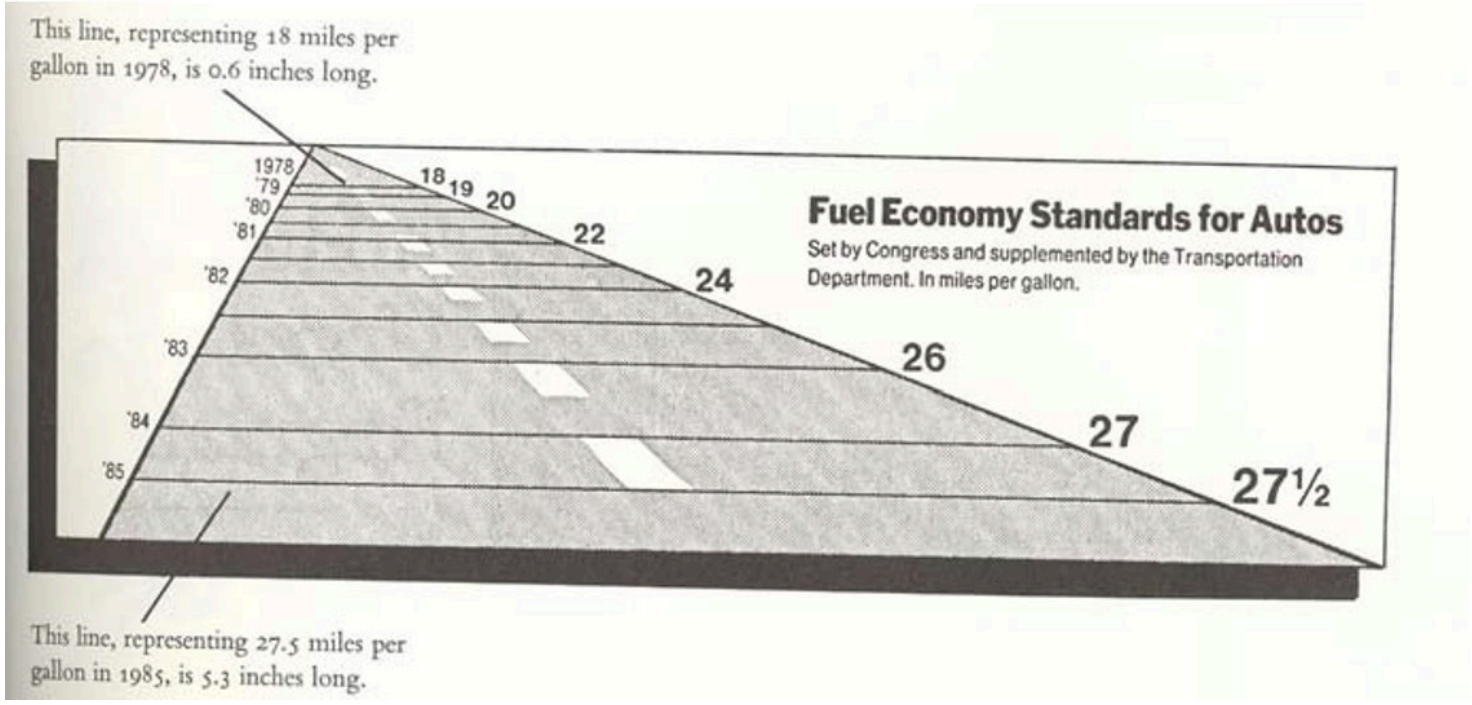



Source: Yau, N. (2012). Fox News continues charting excellence. FlowingData.

# Avoid Distortion: The Lie Factor

$$\text{Lie Factor} = \frac{\text{Size of effect shown in graphic}}{\text{Size of effect in data}}$$




Source: Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). Graphics Press, p. 57.

# Avoid Distortion: Calculating the Lie Factor

**Both "effects" are a percent change:** (new value − old value) / old value. Both sets of numbers are printed on Tufte's figure.

**Effect in the data** 18 mpg in 1978 to 27.5 mpg in 1985: $\frac{27.5 - 18}{18} = \mathbf{0.53\ (53\%)}$

**Effect in the graphic** a line 0.6 inches long to a line 5.3 inches long: $\frac{5.3 - 0.6}{0.6} = \mathbf{7.83\ (783\%)}$

**Lie factor** $\frac{\text{effect in the graphic}}{\text{effect in the data}} = \frac{7.83}{0.53} = \mathbf{14.8}$

**A lie factor of 1 means the graphic is honest.** Tufte treats values above 1.05 or below 0.95 as substantial distortion. This chart shows the change about 15 times larger than it was.

Source: Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). Graphics Press, p. 57.

# Avoid Distortion: A Summary

**Label clearly**

Clear, detailed, and thorough labeling, with appropriate scales.

**Keep the lie factor near 1**

The size of the effect in the graphic should be directly proportional to the numbers it represents.

**Show data variation**

Variation in the graphic should come from the data, not from the design.

Based on Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). Graphics Press, ch. 2.

# Part 2: Theory of Data Graphics

# The Data-Ink Ratio

**Data-ink** is the non-erasable ink used to present the data. If the data-ink were removed, the graphic would lose its content.

**Data-ink ratio** $= \dfrac{\text{data-ink}}{\text{total ink used to print the graphic}}$

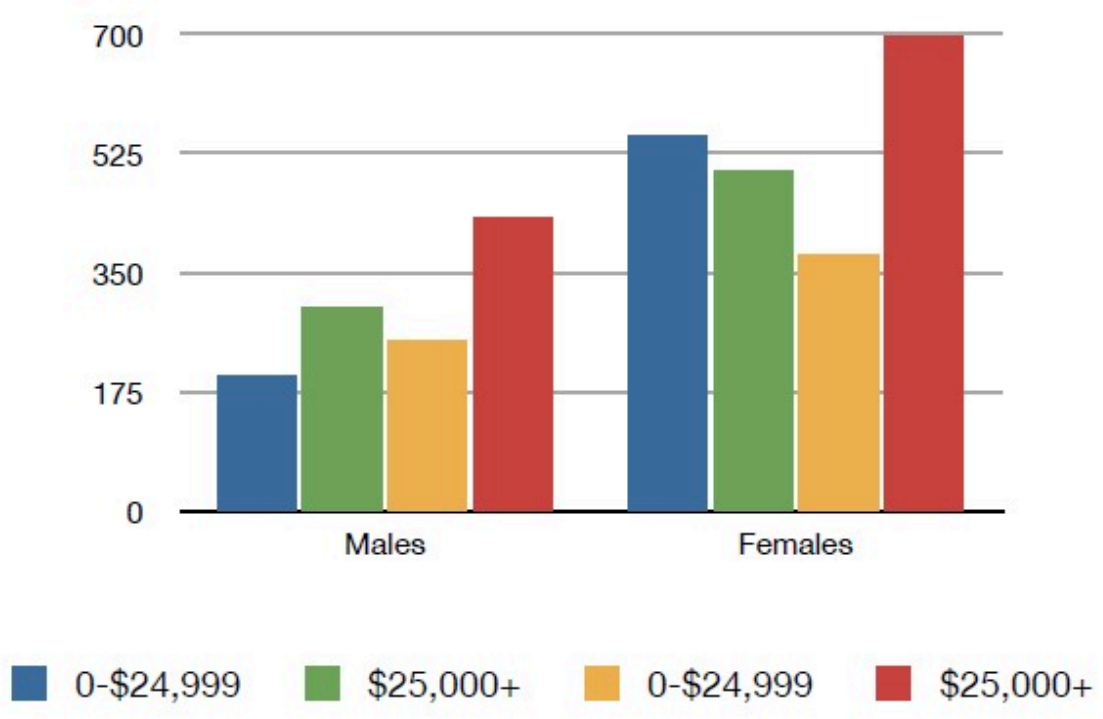
$=$ the proportion of a graphic's ink devoted to the non-redundant display of data information

$=$ 1 − the proportion of a graphic that can be erased without losing data information

Source: Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). Graphics Press, ch. 4.

# The Data-Ink Ratio

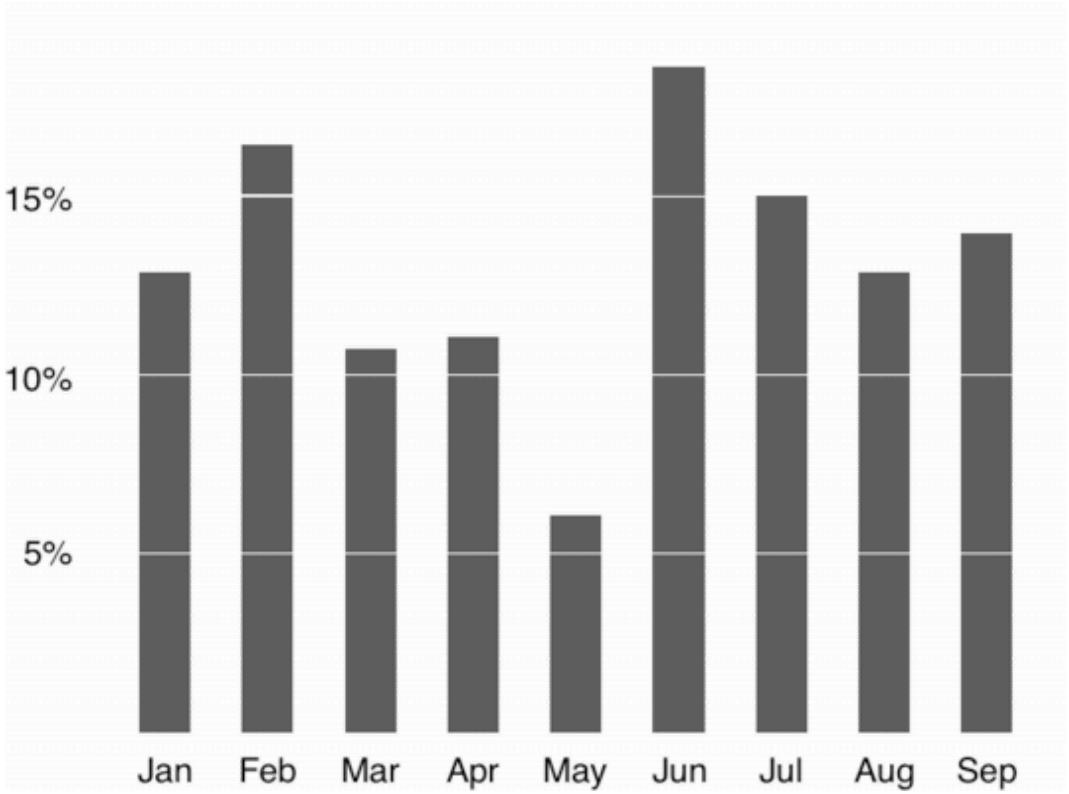
**Data-ink** is the non-erasable ink used to present the data. If the data-ink were removed, the graphic would lose its content.




Source: adapted from Hanspeter Pfister, lecture notes for CS 171, Harvard University. cs171.org

# Focus on Data: Avoid Chartjunk

**Chartjunk is the extraneous visual elements that distract from the message.**




Source: Tim Bray, The Data-Ink Ratio Example (step 6 of 6).

# Other Design Considerations

**Subjective principles (Pat Hanrahan)**

**Aesthetics:** attractive things are perceived as more useful than unattractive ones.

**Style:** communicates brand, process, and who the designer is.

**Playfulness:** encourages experimentation and exploration.

**Vividness:** can make a visualization more memorable.

**Is chartjunk always bad?**

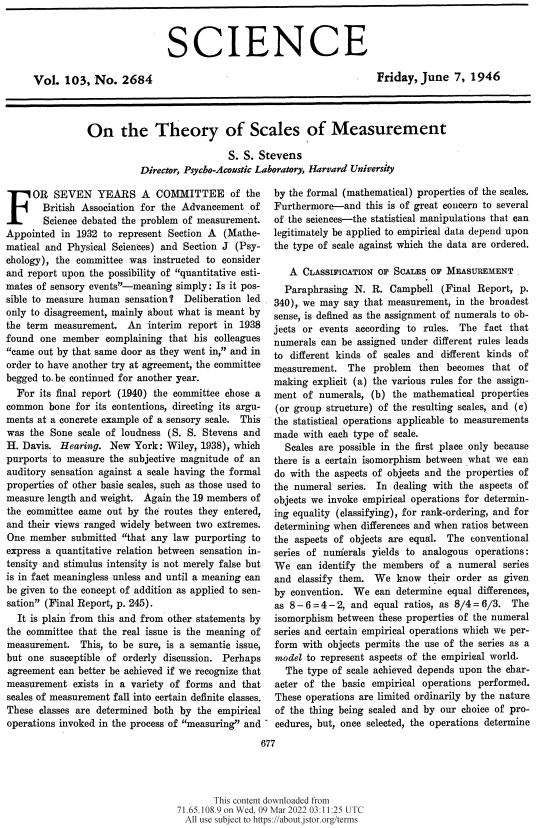
Bateman et al. (2010) compared embellished charts with plain ones. People described the embellished charts **no less accurately**, and **remembered them significantly better** after two to three weeks.

Maximizing the data-ink ratio is a guideline, not a law. Decoration that does not distort the data can help a chart be remembered.

Sources: Pfister, CS 171 lecture notes, based on comments by Pat Hanrahan (2007), discussed at perceptualedge.com/blog/?p=199. Bateman, S., et al. (2010). Useful junk? The effects of visual embellishment on comprehension and memorability of charts. *CHI 2010*, doi:10.1145/1753326.1753716.

# Part 3: Data Types and Visual Encoding

# Stevens (1946): Scales of Measurement



SCIENCE

Vol. 103, No. 2684 Friday, June 7, 1946

## On the Theory of Scales of Measurement

S. S. Stevens

*Director, Psycho-Acoustic Laboratory, Harvard University*

FOR SEVEN YEARS A COMMITTEE of the British Association for the Advancement of Science debated the problem of measurement. Appointed in 1932 to represent Section A (Mathematical and Physical Sciences) and Section J (Psychology), the committee was instructed to consider and report upon the possibility of "quantitative estimates of sensory events"—meaning simply: Is it possible to measure human sensation? Deliberation led only to disagreement, mainly about what is meant by the term measurement. An interim report in 1938 found one member complaining that his colleagues "came out by that same door as they went in," and in order to have another try at agreement, the committee begged to be continued for another year.

For its final report (1940) the committee chose a common bone for its contentions, directing its arguments at a concrete example of a sensory scale. This was the Sone scale of loudness (S. S. Stevens and H. Davis. *Hearing*. New York: Wiley, 1938), which purports to measure the subjective magnitude of an auditory sensation against a scale having the formal properties of other basic scales, such as those used to measure length and weight. Again the 19 members of the committee came out by the routes they entered, and their views ranged widely between two extremes. One member submitted "that any law purporting to express a quantitative relation between sensation intensity and stimulus intensity is not merely false but is in fact meaningless unless and until a meaning can be given to the concept of addition as applied to sensation" (Final Report, p. 245).

It is plain from this and from other statements by the committee that the real issue is the meaning of measurement. This, to be sure, is a semantic issue, but one susceptible of orderly discussion. Perhaps agreement can better be achieved if we recognize that measurement exists in a variety of forms and that scales of measurement fall into certain definite classes. These classes are determined both by the empirical operations invoked in the process of "measuring" and by the formal (mathematical) properties of the scales. Furthermore—and this is of great concern to several of the sciences—the statistical manipulations that can legitimately be applied to empirical data depend upon the type of scale against which the data are ordered.

### A Classification of Scales of Measurement

Paraphrasing N. R. Campbell (Final Report, p. 340), we may say that measurement, in the broadest sense, is defined as the assignment of numerals to objects or events according to rules. The fact that numerals can be assigned under different rules leads to different kinds of scales and different kinds of measurement. The problem then becomes that of making explicit (a) the various rules for the assignment of numerals, (b) the mathematical properties (or group structure) of the resulting scales, and (c) the statistical operations applicable to measurements made with each type of scale.

Scales are possible in the first place only because there is a certain isomorphism between what we can do with the aspects of objects and the properties of the numeral series. In dealing with the aspects of objects we invoke empirical operations for determining equality (classifying), for rank-ordering, and for determining when differences and when ratios between the aspects of objects are equal. The conventional series of numerals yields to analogous operations: We can identify the members of a numeral series and classify them. We know their order as given by convention. We can determine equal differences, as $8-6=4-2$, and equal ratios, as $8/4=6/3$. The isomorphism between these properties of the numeral series and certain empirical operations which we perform with objects permits the use of the series as a *model* to represent aspects of the empirical world.

The type of scale achieved depends upon the character of the basic empirical operations performed. These operations are limited ordinarily by the nature of the thing being scaled and by our choice of procedures, but, once selected, the operations determine



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Source: Stevens, S. S. (1946). On the Theory of Scales of Measurement. *Science*, 103(2684), 677-680.

# Data Types (from S. Stevens, Theory of Scales)

| Scale | Basic Empirical Operations | Mathematical Group Structure | Permissible Statistics (invariantive) |
|---|---|---|---|
| NOMINAL | Determination of equality | *Permutation group* $x' = f(x)$ $f(x)$ means any one-to-one substitution | Number of cases<br>Mode<br>Contingency correlation |
| ORDINAL | Determination of greater or less | *Isotonic group* $x' = f(x)$ $f(x)$ means any monotonic increasing function | Median<br>Percentiles |
| INTERVAL | Determination of equality of intervals or differences | *General linear group* $x' = ax + b$ | Mean<br>Standard deviation<br>Rank-order correlation<br>Product-moment correlation |
| RATIO | Determination of equality of ratios | *Similarity group* $x' = ax$ | Coefficient of variation |

# Data Types: Explained

**Nominal**

Values are = or ≠ to other values.
Apples, bananas, oranges, etc.

**Ordinal**

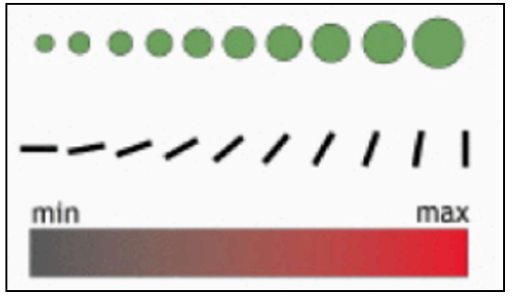
Values obey a < relationship.
Small, medium, and large.

**Quantitative**

We can do math on them.
50 inches, 53 inches, etc.







# Quantitative Data: Interval or Ratio?

**Interval: the location of zero is arbitrary**

Dates (Jan 19), locations (latitude, longitude), temperature in °F.
Only **differences** (intervals) can be compared.

**Test:** is 40°F twice as warm as 20°F? **No.** 0°F is not "no temperature", so the ratio means nothing. The difference of 20°F does mean something.

**Ratio: zero is fixed**

Measurements: length, weight, cost, counts.
The origin is meaningful, so **ratios, proportions, and differences** can all be computed.

**Test:** is $40 twice as much as $20? **Yes.** $0 means no money, so "twice as much" is meaningful.

**Why it matters for charts:** a bar's length shows a value as a distance from zero. That works for ratio data. For interval data, a bar from zero has no meaning, so we use positions (dots, lines) instead.

# Data Terminology

02:00

Task | Your Answer | Answer | Discussion

**Task.** In 2 minutes, identify the data type (nominal, ordinal, interval, or ratio) of each column.

| Order ID | Order Date | Order Priority | Product Container | Product Cost | Ship Date |
|---|---|---|---|---|---|
| 1 | 1/1/2022 | 5 - low | Large box | 25 | 1/5/2022 |
| 2 | 1/4/2022 | 4 - not specified | Small Box | 36 | 1/7/2022 |
| 3 | 1/15/2022 | 2- high | Small Box | 38 | 1/17/2022 |
| 3 | 1/15/2022 | 2- high | Small Box | 41 | 1/17/2022 |
| 3 | 1/15/2022 | 2- high | Jumbo Box | 44 | 1/17/2022 |
| 3 | 1/15/2022 | 2- high | Wrap Bag | 33 | 1/17/2022 |
| 4 | 1/18/2022 | 1- urgent | Small Box | 33 | 1/19/2022 |
| 5 | 1/20/2022 | 3- medium | Medium Box | 40 | 1/23/2022 |
| 5 | 1/20/2022 | 3- medium | Small Box | 16 | 1/23/2022 |
| 5 | 1/20/2022 | 3- medium | Large box | 38 | 1/23/2022 |
| 6 | 1/21/2022 | 1- urgent | Wrap Bag | 7 | 1/22/2022 |

# Data vs. Conceptual Models

**1 Data model**

32.5, 54.0, −17.3, ...

Numbers in a column.

→

**2 Conceptual model**

Temperature

What the numbers mean.

→

**3 Data type we choose**

Continuous to x significant digits: **quantitative**

Hot, warm, cold: **ordinal**

Burned vs. not burned: **nominal**

**The same column can become different data types,** depending on the question we ask. The type we choose decides which visual encodings fit.

# Image Model: Visual (Encoding) Variables

Channels

Position

Size

(Grey)Value

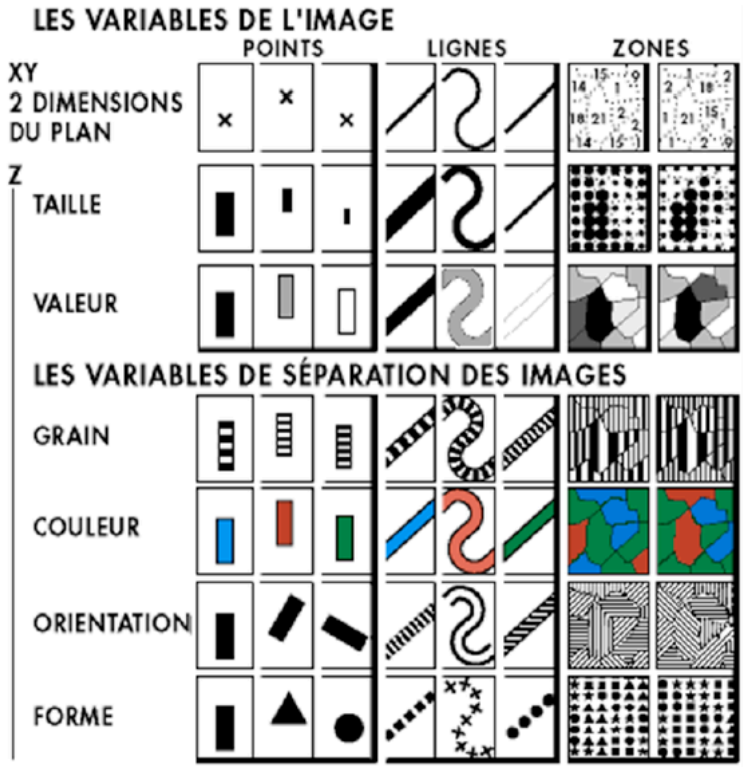
Texture

Color

Orientation

Shape

Marks Points Lines Areas




Source: Bertin, J. (1967). *Sémiologie graphique.* Paris: Gauthier-Villars.

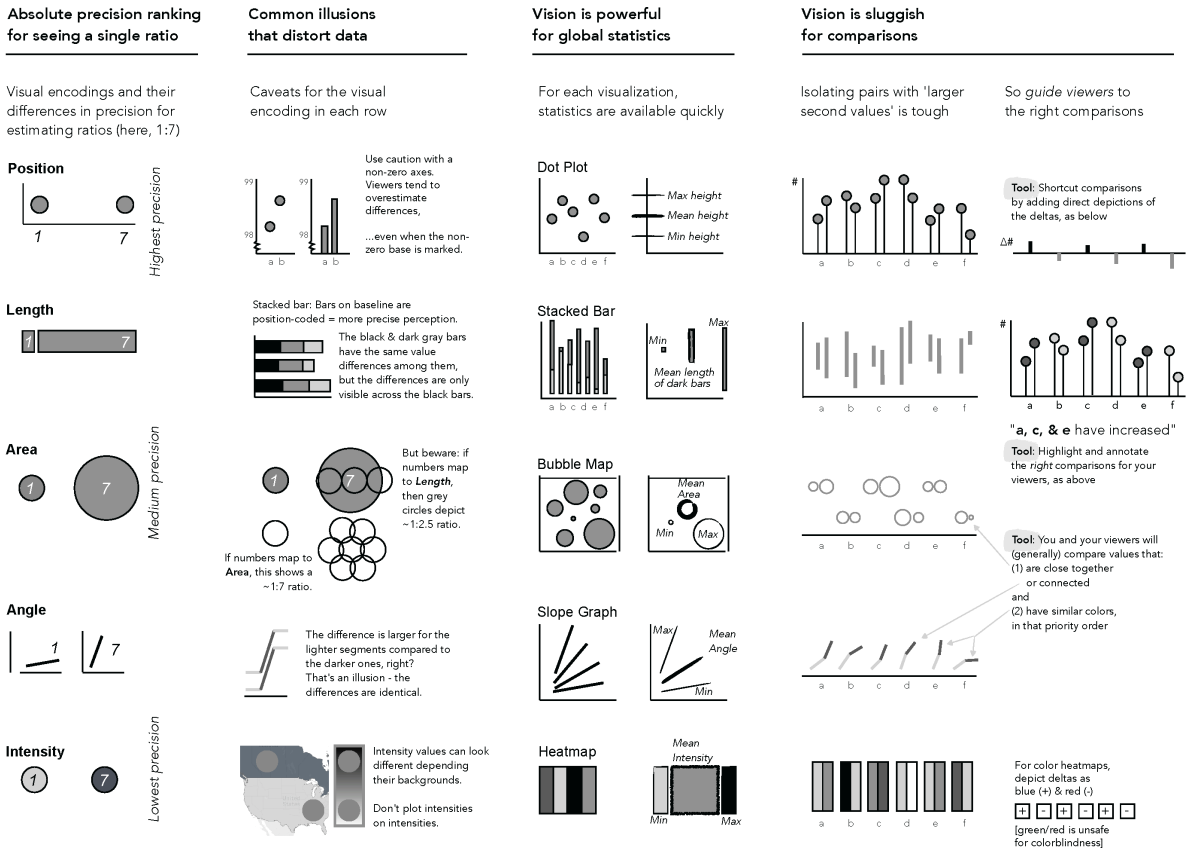
# Mapping to Data Types

| | Nominal | Ordinal | Quantitative |
|---|---|---|---|
| Position | ✔ | ✔ | ✔ |
| Size | ✔ | ✔ | ~ |
| (Grey)Value | ✔ | ✔ | ~ |
| Texture | ✔ | ~ | ✖ |
| Color | ✔ | ✖ | ✖ |
| Orientation | ✔ | ✖ | ✖ |
| Shape | ✔ | ✖ | ✖ |

✔ = Good
~ = OK
✖ = Bad

Source: adapted from Hanspeter Pfister, lecture notes for CS 171, Harvard University. cs171.org

# Visual Channels and Their Precision

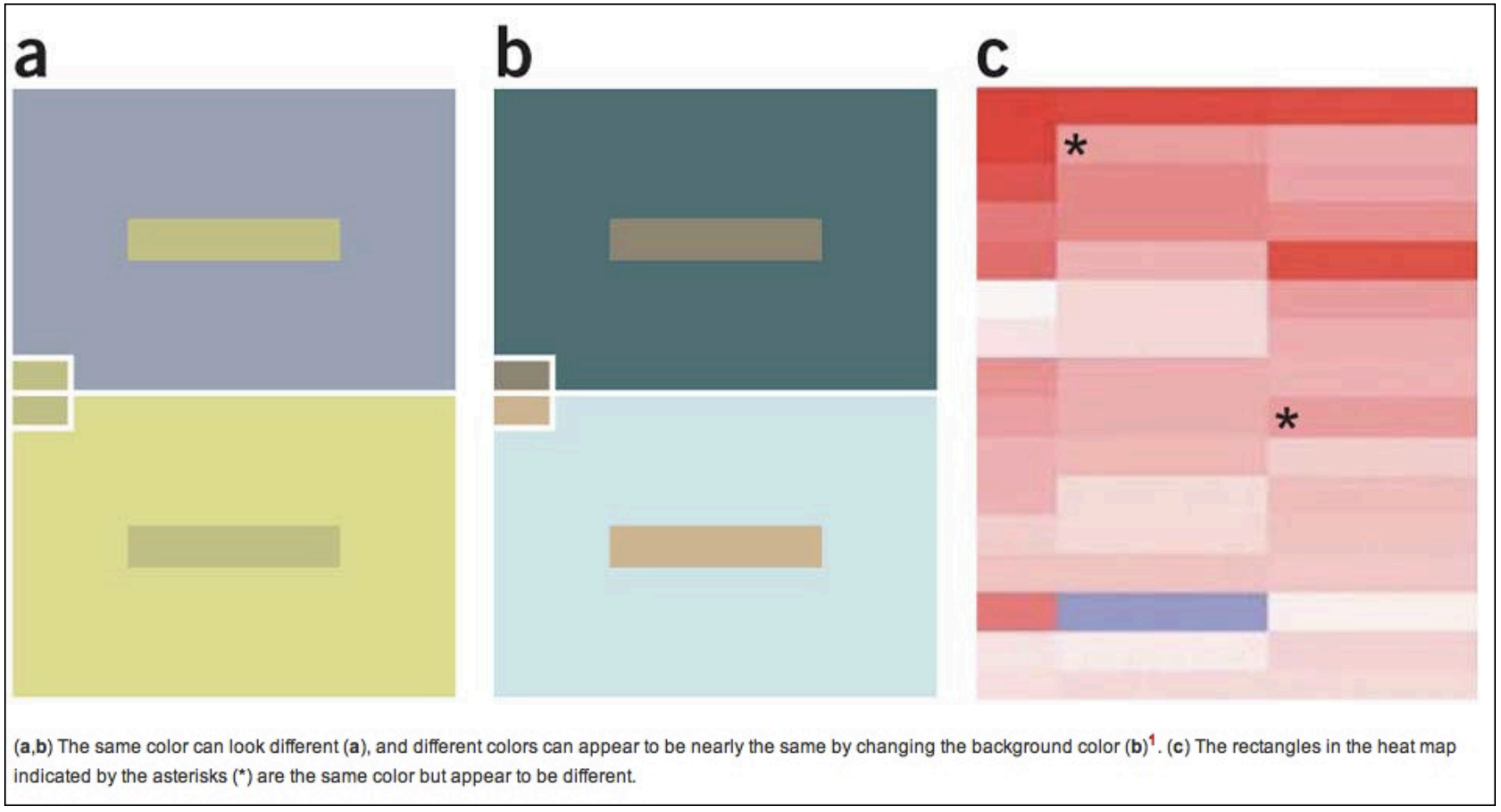



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*Reproduced in Franconeri, Padilla, Shah, Zacks, Hullman (2021). The Science of Visual Data Communication: What Works. Psychological Science in the Public Interest. 22(3):110-161.*

Source: Franconeri, S. L., et al. (2021). The Science of Visual Data Communication: What Works. *Psychological Science in the Public Interest*, 22(3), 110-161.

# Part 4: Color Should Be Used Sparingly

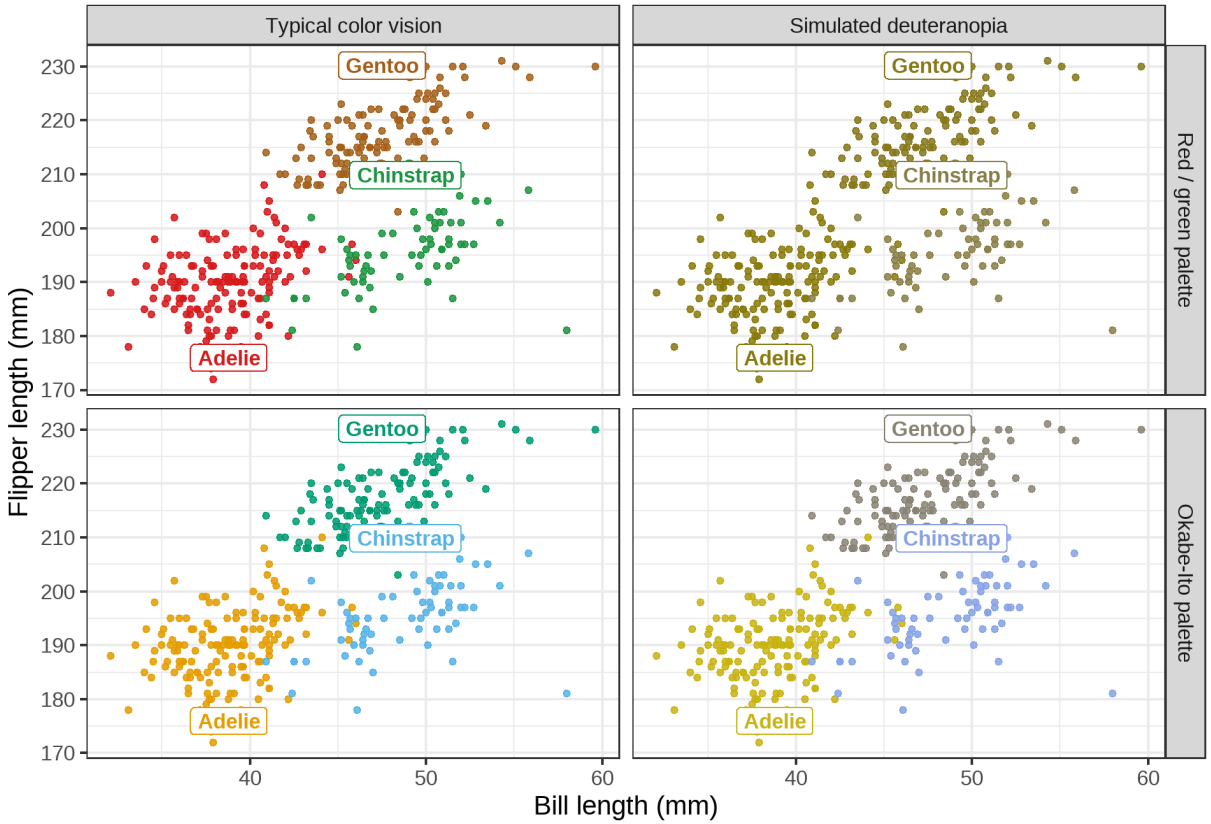
# Simultaneous Contrast Affects Perception




(**a**,**b**) The same color can look different (**a**), and different colors can appear to be nearly the same by changing the background color (**b**)[1]. (**c**) The rectangles in the heat map indicated by the asterisks (*) are the same color but appear to be different.

Source: adapted from Hanspeter Pfister, lecture notes for CS 171, Harvard University. cs171.org

# Color Vision Deficiency




> "About 1 in 12 men have color vision deficiency." (National Eye Institute)

**Top row:** under simulated deuteranopia, the red / green palette turns all three species into similar olive browns.

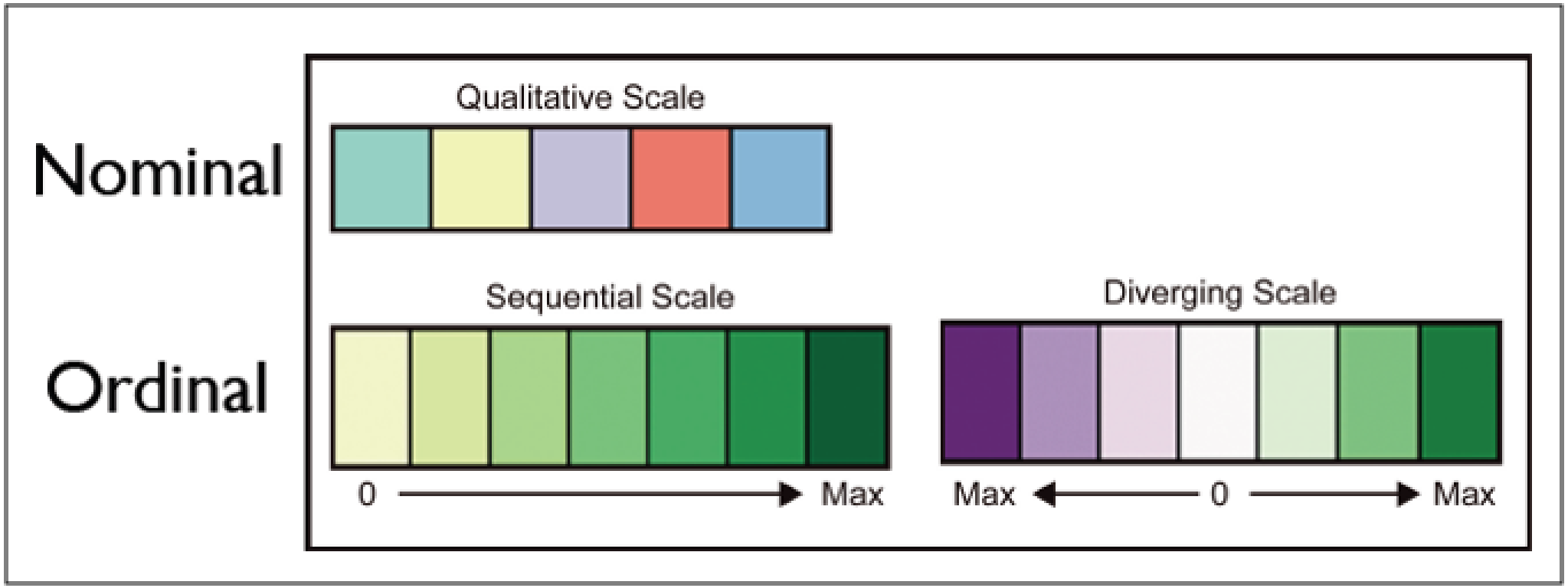
**Bottom row:** the Okabe-Ito palette keeps the three species apart.

**How to accommodate it:** use color sparingly, and use color-blind-safe palettes (e.g., ColorBrewer with "colorblind safe" checked).

Not color blind? This test gives a feel for it.

Data: `datasets::penguins` (Palmer Station LTER). Simulation: `colorspace::deutan()`.

# Color Brewer: Color Scales and Their Selection




**Pick a palette:** colorbrewer2.org lets you choose the number of classes and the scheme type, and filter for colorblind-safe palettes.

Source: Brewer, C. A. (1999). Color Use Guidelines for Data Representation. *Proceedings of the Section on Statistical Graphics*, American Statistical Association.

# Recap

# Summary of Main Points

By now, you should be able to do the following:

- **Explain** Tufte's principles of graphical excellence: show the data, keep the focus on substance, and avoid distortion (Anscombe, Playfair, the data-ink ratio).
- **Select** visual encodings based on data types: nominal, ordinal, interval, or ratio, with position as the channel we perceive most accurately.
- **Apply** color sparingly, choose sequential, diverging, or qualitative scales to match the data, and use palettes that work for color-blind readers.
- **Identify** and critique misleading visualizations, including with the lie factor.

# Before Next Class

**Suggested**

**The Lie Factor and the Baseline Paradox** (de Boer, 2021, Nightingale).
Note what the author means by "baseline", and when the lie factor can mislead for time series and ratio data.

Wilke, *Fundamentals of Data Visualization*: **ch. 4, Color scales** and **ch. 19, Common pitfalls of color use**.
Free online. Chapter 19 includes designing for color-vision deficiency.

**Skim if curious**

**Useful junk? The effects of visual embellishment on comprehension and memorability of charts** (Bateman et al., CHI 2010).
The experiment behind the "Is chartjunk always bad?" card. The paper uses "ratio" differently from how we used it today.

**Monday, Oct 19:** we apply today's principles in code with ggplot2.